Personal Statement

Zhenghao Chen · PhD Application —
Graph Intelligence & Urban Resilience

My path into research began with a lifelong curiosity about geography. As a high
— first clarify the center of a problem, then think across its dimensions
—
and it eventually led me to the conviction that now defines my research: the stru

I entered the Jianxing Honors College at Zhejiang University of Technology, chos
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diffusion equations, and studied deep-learning traffic-flow imputation with RNN
– cloud collaborative aquaculture monitoring system —
the first to apply SlowFast to fish-behavior detection (97.06% accuracy)
—
which was deployed in intensive American-shad farming and later accepted at II
“ Deep Learning for Time Series Forecasting: A Survey ”
(IJMLC 2025), now cited more than 420 times on Google Scholar, which taught

This intuition matured into a systematic study of graph neural networks across t
—
denoising noisy semantic graphs through behavior-conditioned diffusion (IGDM

Beyond model-centric questions, I extend this network perspective to problems w
’ s Infrastructure, AI and Urban Nexus Lab, I quantified how a city ’
s heat accumulation is shaped by the structure of its street network. This grew in
’
s heat-ventilation potential. Linking UTRI to socioeconomic data in Washington,
—
quantitative evidence that heat risk is an environmental-justice issue embedded

These experiences converge on the direction I want to pursue in my PhD. Metho
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mining the phenomena and regularities that connect climate stress, environmen

What I seek in a PhD is an environment where I can work alongside excellent, di

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and where research value can be translated into social value. My starting point f

My goal is to work on the front line of climate adaptation and disaster response,

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I value persistence above all. I hope to carry this thread into climate-change rese